

Detector Design Spec 1 — 4H-SiC Dosimetry Diode (p-n, zero-bias capable)

For: foundry hand-off (ST / FBK / other) From: Petringa group, INFN-LNS — 4H-SiC TCAD study Date: 2026-06-16
Status: process-agnostic physics specification (NOT a mask set)

How to read this document. This is a physics target the foundry maps to its own process — not a mask layout. We do not have ST/FBK design rules, achievable doping windows, or epi grids, so every “fab fills in” cell below is deliberately left to the foundry. The electrostatic targets (doping profile, depletion, full-depletion voltage, CCE-when-depleted) are TCAD model outputs validated against our calibrated 1D twin and C-V data ($R^2 \geq 0.99$, 0 to -50 V). The leakage, breakdown, and zero-bias-efficiency rows are explicitly NOT predictive — see the caveats.

1. Device concept

Planar one-sided p^+/n^- junction on a graded n-type epitaxial layer over an n^+ (or p^+ -substrate, per the telescope convention) wafer, with a guard ring for edge termination. Intended to operate under reverse bias for clinical dosimetry, and to retain a self-powered (zero-bias) mode via the built-in junction field.

2. Electrostatic target spec (TCAD model targets + tolerance — fab maps to process)

Parameter	Target value	Tolerance (fab maps)	Basis
Junction type	p^+/n^- , one-sided	—	design choice
Epi thickness (biased clinical device)	10 μm	$\pm 1 \mu\text{m}$	calibrated config
Graded epi donor profile $N_D(y)$	$N_D(y) = N_{D_bulk} + (N_{D_junction} - N_{D_bulk}) \cdot \exp(-y/L)$	see below	calibrated (Phase 26)
→ $N_{D_junction}$ (near junction)	$2.93 \times 10^{15} \text{ cm}^{-3}$	$\pm 20 \%$	Nelder-Mead C-V fit
→ N_{D_bulk} (deep epi)	$8.82 \times 10^{13} \text{ cm}^{-3}$	$\pm 20 \%$	Nelder-Mead C-V fit
→ L (grading length)	0.99 μm	$\pm 0.3 \mu\text{m}$	Nelder-Mead C-V fit
p^+ anode doping N_A (total)	$1 \times 10^{19} \text{ cm}^{-3}$	fab	active $\approx 1.3 \times 10^{18} \text{ cm}^{-3}$ (≈ 13 % ionized, model)
Junction depth (p^+ emitter)	$\sim 1 \mu\text{m}$	implant/anneal $\pm 0.3 \mu\text{m}$	calibrated config
Relative permittivity ϵ_r	9.7	—	4H-SiC literature
Guard ring	required	geometry per fab design rules	edge termination

Depletion behaviour (calibrated graded profile, verified by TCAD):

Reverse bias	Depletion width W	Note
0 V	1.70 μm	built-in depletion (self-powered drift region)
-10 V	$\sim 9.5 \mu\text{m}$	epi essentially fully depleted ($\sim 9 \mu\text{m}$)
-30 V	(saturated)	already fully depleted; further bias adds field, not depleted volume

The 2D solver ramps reliably to -50 V and reproduces the 1D-twin C-V within 2.6 % across 0 to -50 V. (This is a solver-convergence + C-V-match result; it is not a rated breakdown voltage — see §5.)

Full-depletion voltage V_{fd} (calibrated profile): ~ 4.1 V (5 μm epi), ~ 10.5 V (10 μm epi), ~ 35 V (20 μm epi). Operate at a bias comfortably above V_{fd} for the chosen epi.

Charge collection (reverse bias, TCAD-model, validated vs 1D twin): $\text{CCE} \rightarrow 1$ (> 0.96) once the device is fully depleted. This is a drift-diffusion model result for the simulated p-n junction; not an absolute device guarantee.

3. Zero-bias (self-powered) operation — honest scope

The built-in p^+/n^- junction gives a depletion region $W(0\text{ V}) = 1.70\text{ }\mu\text{m}$ with a built-in field that separates carriers without applied bias — so the device is electrostatically self-powered. However, on the 10 μm clinical epi only $\sim 17\%$ of the epi sits in the 0 V drift field; charge generated deeper collects by diffusion, which our model does not quantify at 0 V. So:

- Defensible now: the device is self-powered; the drift-active 0 V volume is $\sim 1.7\text{ }\mu\text{m}$.
- NOT claimed: a quantitative 0 V collection efficiency on the 10 μm device — that needs transient DD modeling plus one measured zero-bias anchor.

Design lever for a genuinely zero-bias dosimeter (recommended separate variant): $W(0\text{ V}) \propto \sqrt{(V_{bi} \cdot \epsilon / (q \cdot N_D))}$. To get a meaningful depleted fraction at 0 V, use a thinner, lower-doped epi: target $\sim 3\text{--}4\text{ }\mu\text{m}$ epi at $N_D \approx 1\text{--}3 \times 10^{14}\text{ cm}^{-3}$, which depletes $\geq 80\%$ of the epi at 0 V ($W(0\text{ V}) \approx 3\text{--}6\text{ }\mu\text{m}$). This trades signal depth for self-powered operation. The calibrated 2.93×10^{15} junction profile above is the biased clinical device — NOT the zero-bias variant. If a dedicated zero-bias detector is wanted, we will re-run the calibration at the thin/low-doped target and quote its verified $W(0\text{ V})$.

4. Fab fills in (explicitly not modeled here)

- As-grown epi profile, exact peak doping, doping tolerances $\rightarrow$ fab process window.
 - p^+ implant species/dose/anneal and actual activation (we model $\sim 13\%$ ionization).
 - Contact metallization stack, anneal, ohmic contact resistivity, sheet resistance.
 - Passivation, surface preparation.
 - Guard-ring geometry and edge-termination implementation per design rules.
 - Achievable epi-thickness grid and minimum feature sizes.
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5. Caveats — do NOT read these as committed specs

- Dark current / leakage: ESTIMATE ONLY. Our model leakage ($\sim 18\text{ pA}$ at -30 V , 300 K) is a single-point calibration (one fitted trap term), not a first-principles prediction. Use only as a budget estimate; the fab must measure. Do not treat it as a leakage limit/guarantee.
 - No rated breakdown voltage. -50 V is the TCAD solver-convergence envelope; there is no impact-ionization/breakdown model behind it.
 - Edge/guard-ring performance not modeled — our mesh is a laterally-uniform 1D-twin; it cannot predict edge field crowding or termination breakdown.
 - CCE figures are DD-model results for the simulated junction, validated against the 1D twin only.
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6. What is solid (publishable / hand-off-ready)

Electrostatics: built-in potential, depletion vs bias, electric field, C-V ($R^2 = 0.998$ vs data; 2D C-V matches 1D twin $R^2 \geq 0.99$ over 0 to -50 V), the calibrated graded doping profile, full-depletion voltage, and CCE-vs-bias when depleted. These are the dimensioning core and are defensible as model targets with the tolerance bands above.